PSEUDO-COHERENCE IS LOCAL FOR A FINITE COVER BY PRINCIPAL OPEN SETS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Finiteness of a module can be checked on a finite cover by principal open sets. Algebra, Lemma \ref{algebra-lemma-cover}, cited at proof lines 16421 and 16431. Turn finiteness of the localised cohomology modules into finiteness over the ring, and vanish the cohomology above the top degree.
- The m-pseudo-coherent complexes are exactly those with vanishing cohomology above m and finitely generated cohomology in degree m. Lemma \ref{lemma-recognize-pseudo-coherent}, cited at proof lines 16424-16425. Gives the base case of the induction.
- Finiteness of cohomology in the top nonvanishing degree characterises pseudo-coherence one degree at a time. Lemma \ref{lemma-finite-cohomology}, cited at proof lines 16430-16431. Provides finiteness of the top cohomology module after localisation.
- Cone lemma for pseudo-coherence in a distinguished triangle. Lemma \ref{lemma-cone-pseudo-coherent}, cited at proof lines 16439-16440 and 16443-16444. Keeps the cone m-pseudo-coherent after localisation and finishes the induction.

Source reference. Original lemma 066D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7932–7932. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00EO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 542–542. Navigation pointer to a statement the

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Source reference. Original lemma 064W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7880–7880. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7876–7876. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064R: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7875–7875. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $f_1, \ldots, f_r \in R$ be elements which generate the unit ideal. Let $m \in \mathbf{Z}$. Let $K^\bullet$ be a complex of R-modules. If for each i the complex $K^\bullet \otimes_R R_{f_i}$ is m-pseudo-coherent (resp. pseudo-coherent), then $K^\bullet$ is m-pseudo-coherent (resp. pseudo-coherent).*

Proof. We will use without further mention that $- \otimes_R R_{f_i}$ is an exact functor and that therefore

$$H^i(K^\bullet)_{f_i} = H^i(K^\bullet) \otimes_R R_{f_i} = H^i(K^\bullet \otimes_R R_{f_i}).$$

Assume $K^\bullet \otimes_R R_{f_i}$ is m-pseudo-coherent for $i = 1, \ldots, r$. Let $n \in \mathbf{Z}$ be the largest integer such that $H^n(K^\bullet \otimes_R R_{f_i})$ is nonzero for some i. This implies in particular that $H^i(K^\bullet) = 0$ for $i > n$ (and that $H^n(K^\bullet) \neq 0$) see Algebra, Lemma 00EO. We will prove the lemma by induction on $n - m$. If $n < m$, then the lemma is true by Lemma 064W. If $n \geq m$, then $H^n(K^\bullet)_{f_i}$ is a finite R_{f_i}-module for each i, see Lemma 064S. Hence $H^n(K^\bullet)$ is a finite R-module, see Algebra, Lemma 00EO. Choose a finite free R-module E and a surjection $E \to H^n(K^\bullet)$. As E is projective we can lift this to a map of complexes $\alpha : E[-n] \to K^\bullet$. Then the cone $C(\alpha)^\bullet$ has vanishing cohomology in degrees $\geq n$. On the other hand, the complexes $C(\alpha)^\bullet \otimes_R R_{f_i}$ are m-pseudo-coherent for each i, see Lemma 064R. Hence by induction we see that $C(\alpha)^\bullet$ is m-pseudo-coherent as a complex of R-modules. Applying Lemma 064R once more we conclude. □